

Isabel Allum

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EDUCATION

Columbia University

New York, NY

M.S. Computer Science, Machine Learning Track - GPA: 3.997

Expected Dec 2026

Courses: Machine Learning, Reinforcement Learning, Artificial Intelligence, Natural Language Processing

University of Chicago

Chicago, IL

B.A. Economics, Specialization in Data Science - GPA: 3.852

2019–2023

Honors: Cum Laude, Dean's List (2020, 2022)

Courses: Econometrics & Machine Learning, Big Data, Data Science Clinic, Data Construction & Interpretation

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, PyTorch, scikit-learn), R (tidyverse, ggplot2), SQL, Stata

Data Tools: PySpark, Databricks, Git/GitHub, Google Cloud Platform

Methods: Causal Inference (DiD, RD, IV), Regression, Clustering, Tree-Based Models, Time Series Analysis

EXPERIENCE

Computer Science Researcher

June 2026 – August 2026

Columbia University

New York, NY

- Developed a new importance-scoring model for an AI memory system that tracks campus discussions, using five weighted features from social media engagement data, to prevent viral posts from skewing topic rankings.
- Created an evaluation environment for testing formula changes, including score breakdowns and tracking.

Economic Data Associate

Jul 2023 – Jun 2025

Charles River Associates

Chicago, IL

- Conducted statistical modeling and end-to-end data analysis on terabyte-scale datasets using PySpark (Databricks), R, and Stata to support 10+ multi-million dollar antitrust and litigation cases, while mentoring junior analysts.
- Engineered reproducible data validation and feature engineering workflows to generate analysis-ready datasets from raw client transaction data and public economic data.
- Applied econometric methods including regression modeling, panel analysis, robustness checks, and hypothesis testing to quantify competitive effects and evaluate but-for counterfactual scenarios.
- Communicated technical findings to broad audiences through visual exhibits and written reports while ensuring analytical defensibility under legal scrutiny.

PROJECTS

Image Captioning with a Conditioned LSTM

April 2026

- Implemented an image-conditioned LSTM caption generator in PyTorch, fusing pretrained image embeddings with token embeddings at each decoding step to generate natural-language captions.
- Utilized a beam search decoding algorithm to improve caption quality over greedy decoding.

Dinner Party Planner Web Application

Sep 2025 – Dec 2025

- Designed and implemented a normalized PostgreSQL database schema to efficiently store and query users, events, recipes, and ratings across multiple related tables.
- Built a Flask web application deployed on Google Cloud Compute Engine enabling real-time user interaction using SQL-backed recipe search, rating updates, and event management.

Perpetual (Data Science Consulting)

Jan 2023 – Mar 2023

- Co-developed a weighted k-means spatial optimization model in Python to identify efficient distribution and collection hubs for reusable foodware across two cities; presented recommendations to nonprofit leadership.
- Integrated Google and Yelp API data, performed feature engineering, and visualized geographic demand.